AI-Based Loan Approval Prediction System

Complete Project Workflow & System Design

Project Type: Machine Learning + Web Application

Technology: HTML, CSS, JavaScript, Python, Flask, Pandas, NumPy, Scikit-learn, SQLite

1. Problem Statement

Loan approval depends on multiple applicant and loan-related factors such as income, credit history, loan amount, employment status, education and property information. This project proposes a machine-learning web application that predicts whether a loan application is likely to be approved or rejected using patterns learned from historical loan data. The result is an academic prediction and is not a real lender decision.

2. Objectives

• Collect and preprocess historical loan application data.

• Analyze factors associated with loan approval outcomes.

• Train and compare suitable classification algorithms.

• Select and save an appropriate trained model.

• Develop a user-friendly web interface for loan applications.

• Integrate the trained ML model with a Flask backend.

• Display the prediction clearly to the user.

• Test and prepare the application for deployment.

3. Complete System Workflow

Step	Process	Description
1	Open website	User accesses the Loan Approval Prediction application.
2	Application form	User enters applicant and loan details.
3	Validation	System checks required fields and valid values.
4	Preprocessing	Categorical values are encoded and numerical data is transformed as required.
5	ML model	Processed input is passed to the trained classifier.
6	Prediction	Model predicts the likely approval outcome.
7	Result	Website displays the model prediction and supporting information.
8	History (optional)	Prediction records may be stored in SQLite.

4. Machine Learning Workflow

1. Dataset collection

2. Data cleaning and missing-value handling

3. Exploratory Data Analysis (EDA)

4. Feature selection

5. Categorical encoding and numerical preprocessing

6. Train-test split

7. Train multiple classification models

8. Evaluate using accuracy, precision, recall and F1-score

9. Select the most suitable model

10. Save the trained model as a .pkl file

11. Integrate the model with Flask

5. Proposed ML Models

Model	Purpose
Logistic Regression	Baseline classification and interpretable benchmark.
Decision Tree	Tree-based classification with understandable decision rules.
Random Forest	Ensemble classifier suitable for comparison and nonlinear patterns.

6. Example Input Features

Feature	Example / Type
Gender	Male / Female
Married	Yes / No
Dependents	0 / 1 / 2 / 3+
Education	Graduate / Not Graduate
Self Employed	Yes / No
Applicant Income	Numerical amount
Co-applicant Income	Numerical amount
Loan Amount	Numerical amount
Loan Term	Number of months
Credit History	Historical credit indicator
Property Area	Urban / Semiurban / Rural

Note: Final features will be finalized from the selected dataset and must match the trained model.

7. System Architecture

Layer	Technology	Responsibility
Presentation	HTML / CSS / JavaScript	Forms, validation, navigation and result display.
Application	Flask / Python	Receives requests and communicates with the ML model.
Preprocessing	Pandas / Scikit-learn	Transforms input into the model's expected format.
ML	Trained classifier	Produces the loan prediction.
Storage	SQLite (optional)	Stores prediction history for demonstration.

8. Website Pages

• Home Page – project introduction and navigation.

• Loan Prediction Page – applicant and loan input form.

• Result Page – displays the model prediction.

• About / How It Works – explains the project and ML workflow.

• Prediction History – optional SQLite-backed history.

9. Project Folder Structure

LoanWiseAI/
■■■ datasets/loan.csv
■■■ models/loan_model.pkl
■■■ templates/index.html
■■■ templates/prediction.html
■■■ templates/result.html
■■■ static/style.css
■■■ static/script.js
■■■ train_model.py
■■■ app.py
■■■ requirements.txt

10. Four-Day Development Plan

Day	Tasks	Deliverable
Day 1	Dataset, cleaning, EDA, preprocessing, train and compare ML models.	Clean dataset + selected model.
Day 2	Build frontend pages, form, navigation and responsive UI.	Working website interface.
Day 3	Build Flask backend and integrate preprocessing + model.	End-to-end local prediction system.
Day 4	Testing, bug fixing, optional history, deployment preparation, screenshots and presentation.	Final project documentation.

11. Expected Output

The user enters loan application information through a web form. Flask validates and preprocesses the input, passes it to the trained model, and displays a prediction such as **Likely Approved** or **Likely Rejected**. Any probability shown will be clearly identified as a model output, not a guarantee.

12. Limitations and Disclaimer

The system is an academic machine-learning demonstration. Predictions depend on the training dataset, preprocessing, selected features and model limitations. The system must not be represented as a real banking approval engine or as a guarantee of lending decisions.

13. Future Enhancements

• User authentication and dashboards.

• Prediction history and downloadable reports.

• Hyperparameter tuning and additional algorithms.

• Model explainability for influential input factors.

• Cloud deployment.

• Improved security, validation and accessibility.

Submission Project Title: AI-Based Loan Approval Prediction System